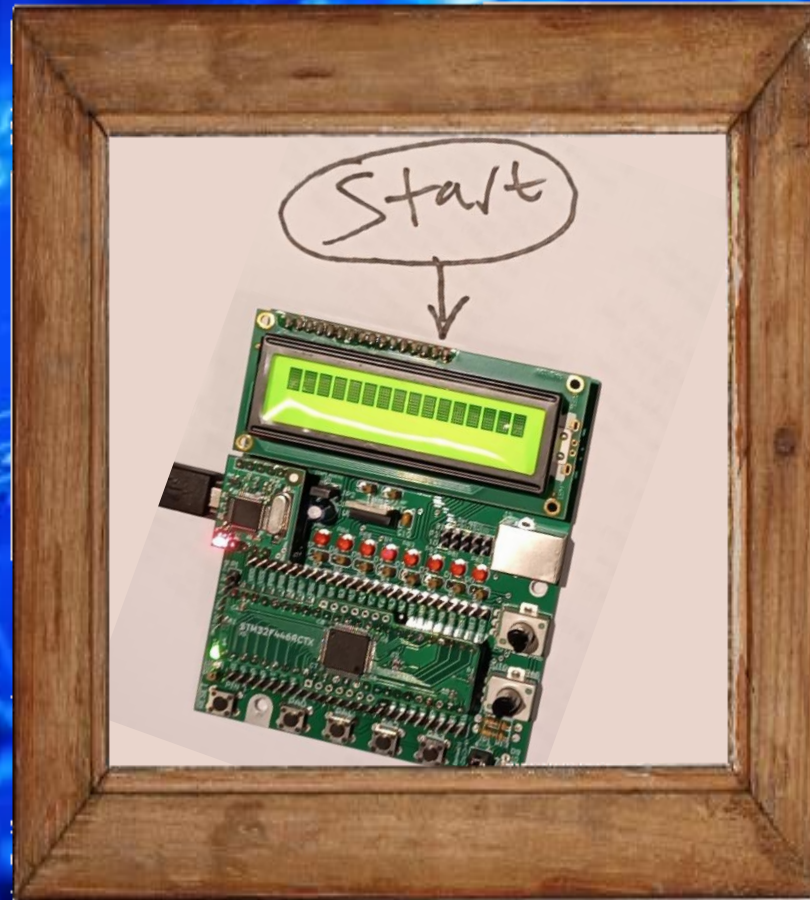


EEE3096S



**Embedded
Systems II**

Welcome! to



Embedded Systems II

Lect. L1

Course

Prof. Simon Winberg

Lecturers:

Mr. Best Nkhumeleni



Electrical Engineering
University of Cape Town

Intro...

```
[simon@localhost]$ whoami
```

```
|
```

Hello! & whoami

Greetings students!

I'm

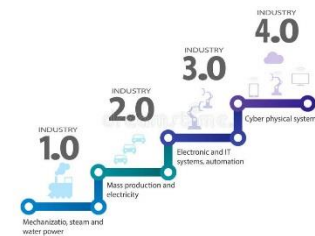
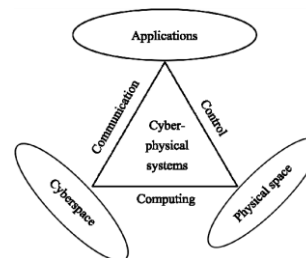
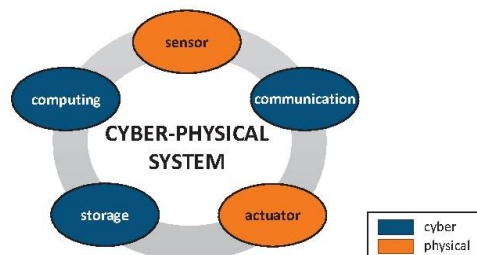
Simon Winberg

And I'm passionate about

Embedded Systems

~~and~~ more correctly

CYBER-PHYSICAL SYSTEMS*



* https://en.wikipedia.org/wiki/Cyber-physical_system

TEACHING SUPPORT STAFF

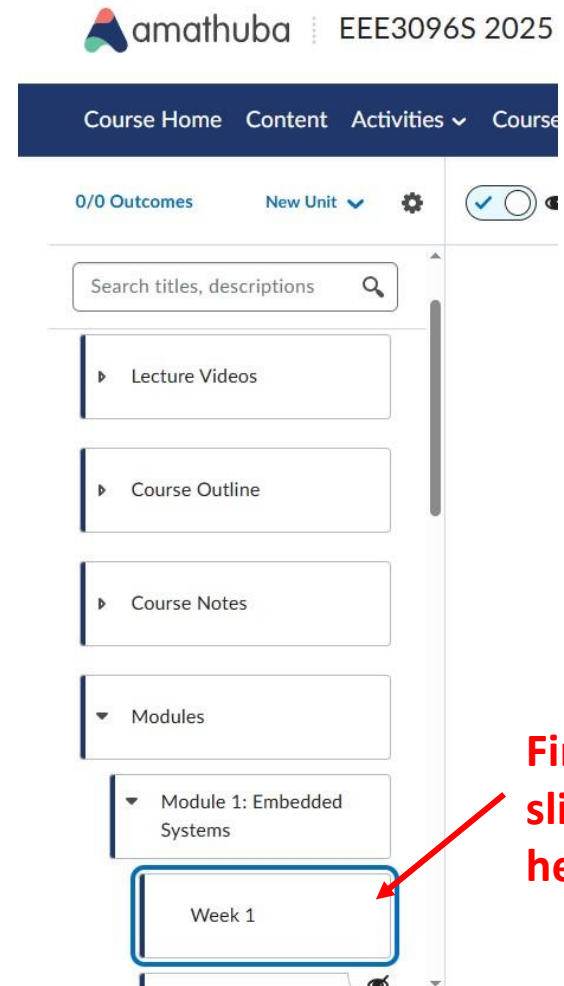
- Course Lecturer:
 - Simon Winberg
 - Best Nkhumeleni
- TAs:
 - Travimadox Webb

Tutors will be available in the prac sessions and some allocated to online assistance

Overview of the learning process

Approach to learning:

- The course is divided into multiple modules, each takes a week or two.
- The lecture slides will be on Amathuba in Content -> Modules :
- The course is divided into multiple modules, each takes a week or two (two-week having 'A' and 'B' parts)



Overview of the learning process

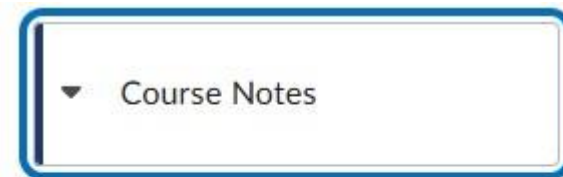
Approach to learning (cont.):

- Tuesday lectures start by introducing topics and planned activities for the week.
- **Q&A** or discussion at the end of each lecture session
- Class **Activities / Short Quizzes**
 - to encourage learning, add interaction
 - various class activities, may need use of phone/tablet (often can be done in small team with discussion)
- **Pracs:**
 - New prac start about every second week
 - Venue: Red Lab (default)
 - Tutors available during scheduled lab sessions

Course Notes

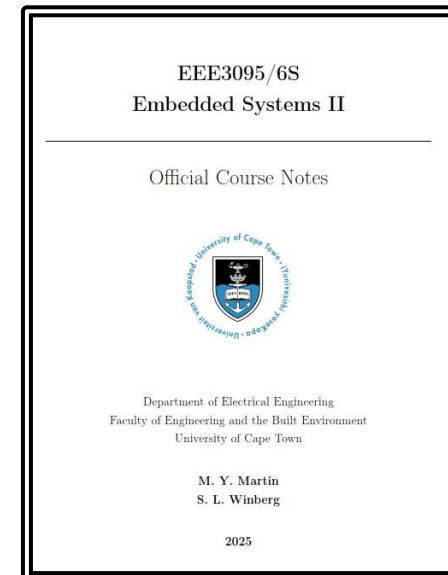
Course Notes:

- Has **additional detail** & written elaboration related to learning topics.
- **Supporting known-how** and concepts (helps pracs, coding in C, and comprehension)
- Lectures point to parts of the course notes
- Note: the lectures include aspects beyond course notes, connecting with concepts and theories beyond technical principles and know-how



EEE3096S 2025
Course Notes

**This thing
to read**





Start of Learning Topics

Week #1: Setting the Context

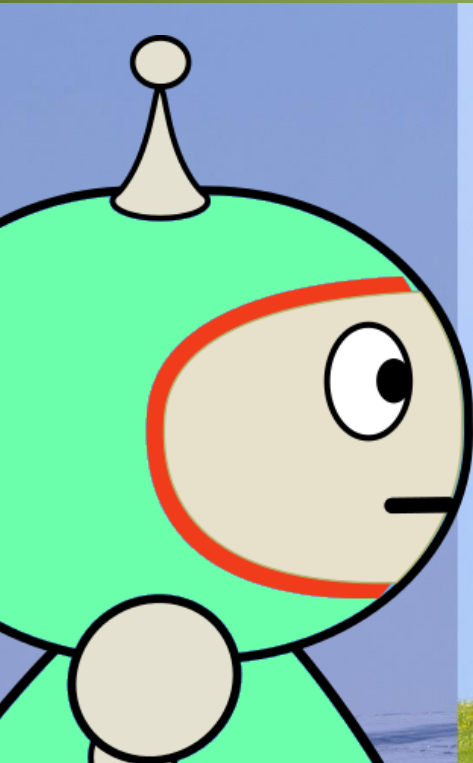
Lessons Plan

- ✓ Introductions
- ✓ Learning approach
- Introduction to ES II and CPS
 - The changing landscape of work: Industry 4++ (setting the flavour)
 - ES jobs & what employers want ... and reason for our curriculum
- Important terminology and concepts
 - Important terminology. Some history and the recent concepts of Cyber-Physical Systems (CPS), 4IR, internet of things (IoT), and how these related to embedded systems.
- Terminology and concept basics
 - What are real-time systems?
 - Hard and Soft Deadlines
 - Optimization of Embedded Systems.
 - SOCs and their benefits
 - Security considerations
 - Deployment mechanisms
 - IoT Networks (and why these are so hot)

Lecture 1 – Intro to ES II

- **Changing landscape of work:**
things we should know... and why we have this course
- What this course involves
- What is...
 - What is an embedded system? (recap)
 - What is a cyber-physical system?
 - What is ubiquitous computing?
- Why combine these in this course?
- All aboard the Embedded Systems Express
- Class business:

Amathuba site & resources; Important issues ...
Getting ahead in EEE3096/5S.



The Changing Landscape of Work

And why this could be good to know...

I like to start a course with some big thinking perspectives – before we go directly into the specific development and low-level issues

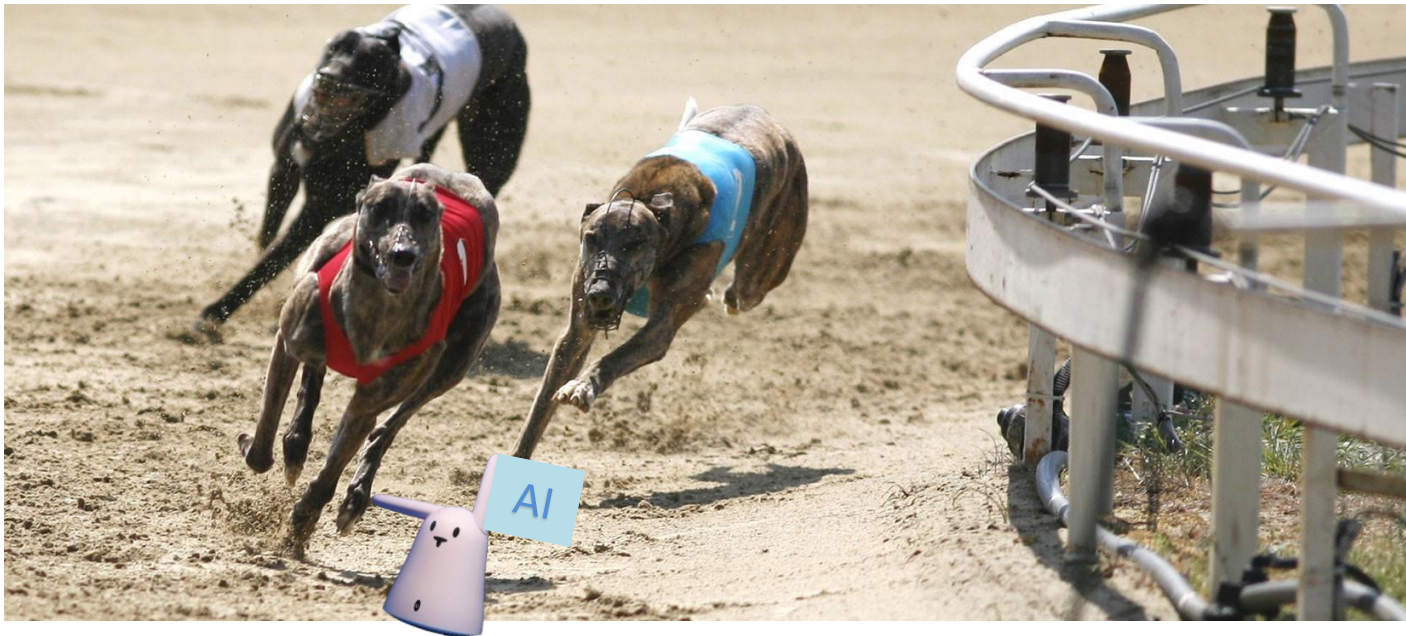
What are they thinking or planning?

Industry 4.0

Has been about for a bit...

‘Now new games are afoot!’*

** (an adaptation of Shakespeare's "King Henry V")*



Industry 4 has been running a while!

Now: **Industry 5** – Human & Machine collaboration

Industry 6 AI, quantum computing & beyond (not fully here?)

Issue of scope

This course doesn't really get much beyond Industry 4.0 concepts (and just a little of it)

(regardless of how much I'd like to get into more ML, AI and Quantum Computing (QC) technologies to assist or apply in the context of embedded systems and sensors development 😊).

NOW BACK TO INDUSTRY 4.0 ...

So... What is Industry 4.0 & 5.0?



We know!!!

... or do you?

Industry 4.0

Industry 4.0 =

the name for the current approach in automation and data exchange in manufacturing technologies.

This includes, in particular, harnessing the power of:

- Cyber-physical systems
- The Internet of things
- Cloud computing
- Cognitive computing

Leading to the concept of
“smart factories”*



Smart factories are at the center of the Industry 4.0 (connected industry) revolution.

Many of these plans are already in place, especially at large companies that want to stay ahead of the ‘game’.

** Which connects increasingly with 5-IR*

Industry 4.0

... harnessing the power of:

- Cyber-physical systems
- The Internet of things
- Cloud computing
- Cognitive computing

*Look these terms up if you
aren't sure what they mean!!*

Obviously this can – and has – lead to **massive changes** in **the way production works...** and more generally how **modern businesses will work**. These concepts **are** reaching beyond 'factories', making more widespread changes in the way we do things.

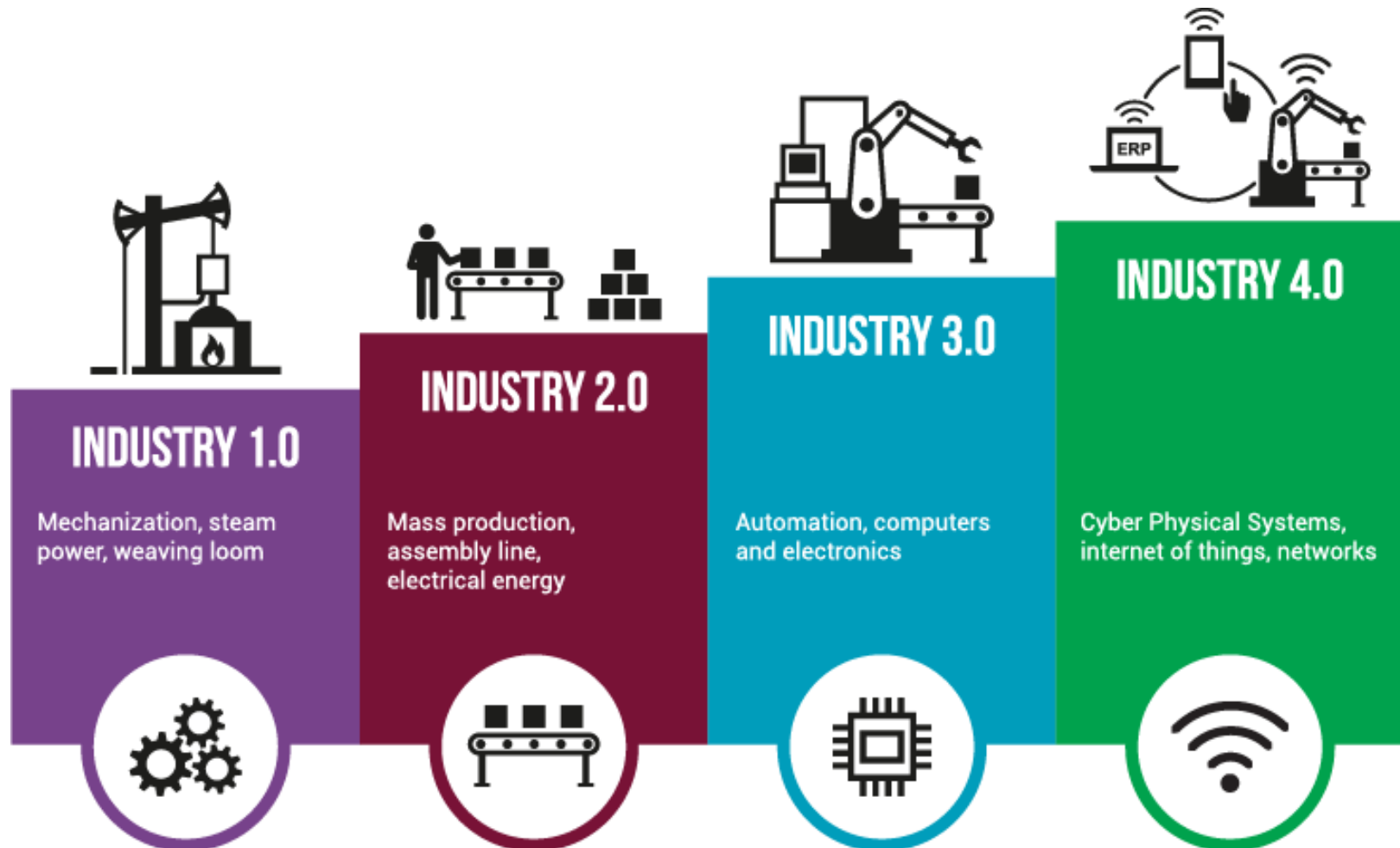
Industry 5.0

... harnessing the power of:

- **Beyond cyber-physical systems**, cloud computing, etc.
- Acknowledging **needs for humans** 😊 (to some extent)
- Leveraging advanced technologies, such as AI and robotics, for collaborative enhanced manufacturing and other industrial processes.
- Builds on Industry 4.0 by emphasizing human-centricity, sustainability, and resilience
- Moving beyond purely efficiency-driven goals.

Obviously this can – and has – lead to **massive changes**. Has a reach far beyond ‘factories’, it is making more widespread changes in the way we do things.

'Industry 4.0' and its enabling technologies.



'Industry 4.0' is a term coined at the Hannover Fair in Germany, 2011, to describe how organizations are changing*

Image source: Nitin's Fundas Blog Spot – "Reflections on Leadership, Innovation, Self Development, Spirituality and Transforming India" (<http://nseth71.blogspot.com/>)

* K. Schwab, "The Fourth Industrial Revolution", Portfolio Penguin. 2017

Each advance has required
More skilled workers

Education



Generally, an ever-greater portion
of the workforce being
‘knowledge workers’*

But not just... Industry 5 and 6 some differences: heightened human-centric methods, teamwork, eco-awareness, willing to learn, work with people & AI.

* Peter Drucker (1959) “The Landmarks of Tomorrow”

The Industry 5 or 6 situation

Undesirability of the: 'Rote Proceduralist'

Desirability of the: Shift from Knowledge to Human-Centric Skills

5IR^[1] Most desired individuals possessing skills that machines don't have:

- **Creativity and Innovation:** novel solutions to complex problems.
- **Critical Thinking:** Questioning, analyzing, and improving things.
- **Emotional Intelligence:** Collaborating effectively, leading with empathy
- **Adaptability:** Learning and thriving in a constantly changing technological and social landscape.

[1] Sneha, V., & Kavitha, R. (2024). Exploring the economy of creativity and culture in the light of Industry 5.0

IS Industry 4.0

GOOD

OR

BAD

??



thinking piece

Are Industry 4.0 Technologies

You might ask the same of the previous industrial revolutions...



GOOD

or

BAD

?



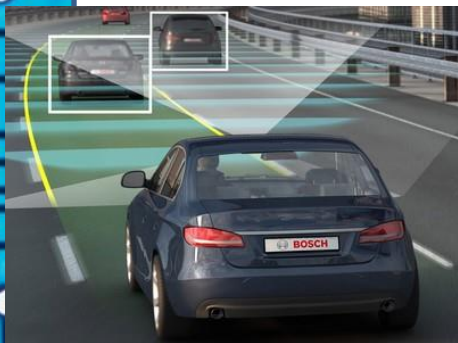
e.g. Industry 1.0 was epitomized by machination, machines people used to work faster. But it demanded a huge number of people who knew how to use these machines. It was a big transition from serve poverty and poor health towards people who essentially had more freedom.

Technologies that are 'successful' may impact the world to a greater or lesser extent. But generally once a technology is invented it can't be undone ('immutability of invention').

So you might as well see if it can benefit you and others 😊

Impact on jobs (slight aside)

- Major impacts are:
 - **Innovation** at the forefront like never before
 - **Speed**: things happening faster than ever
 - **Labour substitute**: repetitive and highly prescribable tasks done by machines...



Think: taxi drivers vs.
self-driving cars

Think: human soldier vs.
The Atlas and similar Robot Soldiers

Think: cashiers
vs. self-checkout
/ Tesco scan as
you shop



What jobs most vulnerable (to 4IR)?

Put very briefly...

Jobs most prone to automation*

Least prone to automation

Probability	Occupation	Probability	Occupation
0.99	Telemarketers	0.0031	Mental Health and Substance Abuse Social Workers
0.99	Tax preparers	0.0040	Choreographers
0.98	Insurance Appraisers, Auto Damage	0.0042	Physicians and Surgeons
0.98	Umpires, Referees, and Other Sports Officials	0.0043	Psychologists
0.98	Legal Secretaries	0.0055	Human Resources Managers
0.97	Hosts and Hostesses, Restaurant, Lounge, and Coffee S	0.0065	Computer Systems Analysts
0.97	Real Estate Brokers	0.0077	Anthropologists and Archaeologists
0.97	Farm Labor Contractors	0.0100	Engineers/ Technologists**
0.96	Secretaries and Administrative Assistants, Except Legal, Medical & Executive	0.0130	Sales Managers
0.94	Couriers and Messengers	0.0150	Chief Executives

← for 5-IR, not as much

You Are Here

Where do (elec./comp.) engineers and computer scientists end up?...

(actually, new understanding showing much need and opportunities for human-centric jobs)

* K. Schwab, "The Fourth Industrial Revolution", Portfolio Penguin. 2017

**The author more indicates marine engineers and architectures in the original table but reading further in the book you can interpret that engineers, particularly those involved in developing and maintaining complex systems are unlikely to become automata, at least not anytime soon.

Should we be worried?

About **4IR** impact on engineers (our engineering / science areas)

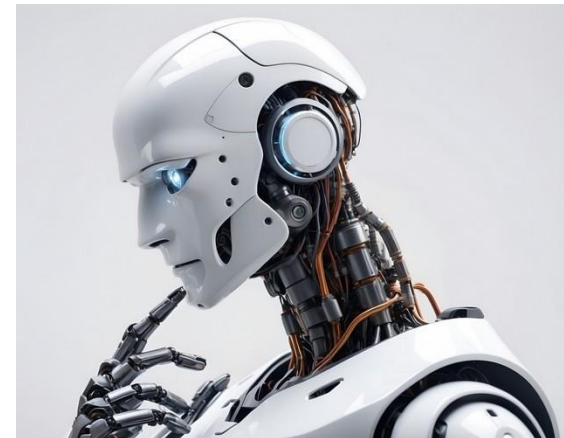
...

PROBABLY NOT A PROBLEM FOR US! 😊

About AI and other **5IR** and beyond approaches? ...

I have no certain answer (yet)

ChatGPT has some good insight on this though...
but maybe you don't want to get distracted on that yet!! There is potentially BIG changes... and (sorry, bit overexcited) but computing (including ES), mechatronics, electronics, energy systems and management, among heightened needs.



So, the short answer* is...

We probably needn't be worried! 😊

But hopefully this little journey though industry trends and how embedded systems and CPS are strongly related to methods of many modern business has been informative, if not inspiring.

And remember that

“Education is the most powerful weapon which you can use to change the world.”

... said who? ... — Nelson Mandela

And hopefully you will use what you learn to make the world a better place!

* To the question of if we are going to be automated anytime soon ...

ES Career Thoughts & ES curriculum design choices

See further optional extra
reading from slide 45



next 3 slides give some insight into the chosen
curriculum design, hopefully clarifies approach used

Embedded Systems Jobs & What Employers Want

... and reasoning for the ES2 and ES3 curriculum choice.

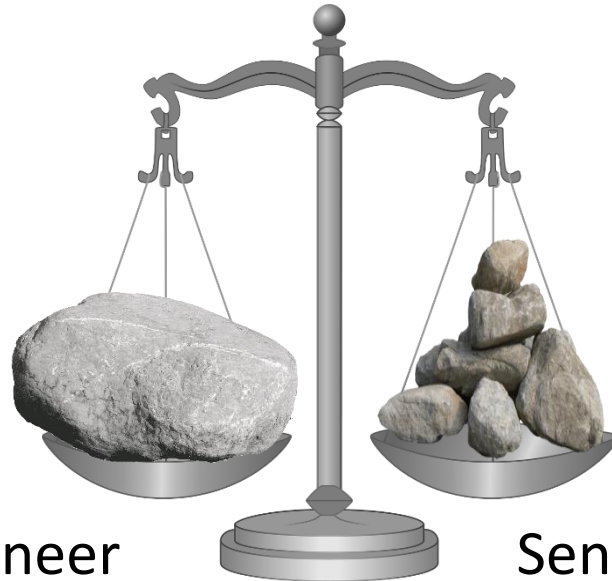
We have a flavour of **training towards Professional ES Engineers**... touching on what such professionals should know. But the main focus in ES2 is on development, predominantly implementation also highly relevant to other fields, such as electrical engineering, telecoms, mechatronics, and C S.

- ❑ Certainly, **professional ES engineers are expected to know**, and have experience, with being able to do the **lower-level development tasks**, such as:
 - Coding, selecting components, knowing about PCB design, comms, hardware/software interfacing, installing and using embedded O/S
- ❑ But also **expected to know the high-level aspects**, which obviously imply lower-level tasks but which other members of the team may do, e.g.:
 - Requirements analysis, specification, high-level design for both software and hardware aspects, certification needs, ethics, inspecting procedures, skills needed for a projects, consultants, (team) leadership, engaging clients and users, documentation.

Embedded Systems Jobs & What Employers Want

Often fewer, more focused and clearly defined tasks to do (under supervision of senior engineer).

Newbie Engineer



Often more, but less well-defined tasks that are done (often in close collaboration / review of specialists, e.g. certification officer or discipline expert).

Senior Engineer

Different companies or organizations may have quite different approaches to what their engineers do, where they are involved, how they progress at the company (hence a disclaimer that this view is not universal).

Often, though, newbie (or junior) engineers are recruited to fill a particular need. E.g. needing someone to do coding, or help with PCB design. Often responsibilities shift over time, as management becomes more aware of an individual's expertise and interests; and identifies particularly dependable people.

I use a balancing scaling to illustrate this: that initially one might have fewer but bigger jobs. With more seniority there are likely more things to keep tabs on, and (hopefully) colleagues helping to get them done. The scale emphasizing how employers should (ideally) be aware of their employees needing a work-life balance, limits to what one person can do.

Rocks Image source: <https://pngimage.net/piedras-png-1>

Embedded Systems Jobs & What Employers Want

... and reasoning for the ES2 and ES3 curriculum choice.

Curriculum reasoning

Students should have options to do some of their learning as

- *'surface learning'* : to see more breadth of the discipline, to know about areas they may want to learn more about, or of areas where there are experts, this may influence their decisions later in their academic studies or in their workplace... and some aspects as
- *'deep learning'* : to greater understanding into particular or specialized aspects of the discipline.

Keeping in mind that an expert, e.g. great at developing device drivers (which would involve deep learning), would still know about the other areas, like there being specialists in signal processing, in communications, etc., that they may influence.



OK, enough about how useful ES is, industry trends and ES skills needs ...
Now we are ready to begin the journey proper!



ALL ABOARD THE EMBEDDED EXPRESS

Onwards to Embedded Systems!

What is an Embedded System?

Defining an Embedded System

“Dortmund (2011) Definition”: [Peter Marwedel]

Embedded systems are information processing systems embedded into a larger product.



“Berkeley (2004) Definition”: [Edward A. Lee]

Embedded software is **software integrated with physical processes**. The technical problem is managing **time** and **concurrency** in the computational systems.



“Brisbane, Australia (2009) Definition”: [John Catsoulis]

An ES is a task-specific computer, which is built into a larger system for the purpose of controlling and monitoring the larger system.



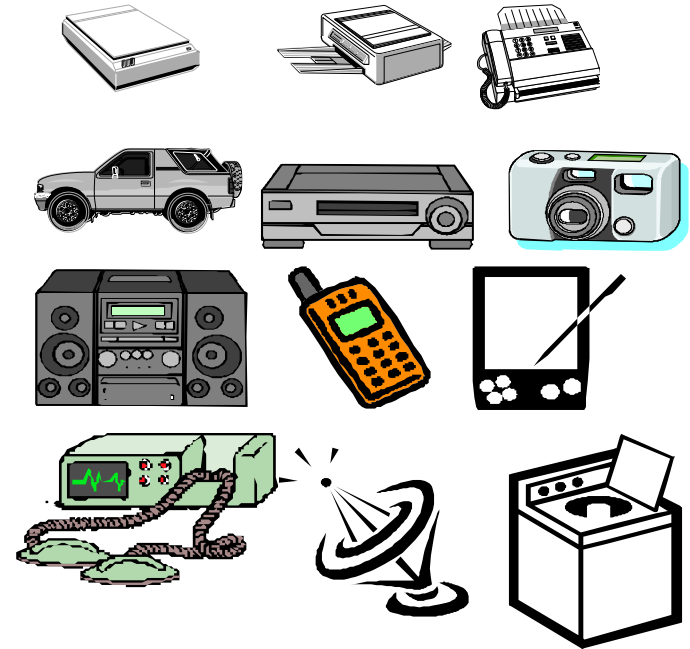
Cyber-Physical (cy-phy) Systems (CPS) are integrations of computation with physical processes [Edward Lee, 2006].

Cyber-physical system (CPS) = Embedded System (ES) + physical environment →

Some Embedded Systems

Anti-lock brakes
Automatic teller machines
Automatic toll systems
Automatic transmission
Avionic systems
Battery chargers
Camcorders
Cell phones
Cell-phone base stations
Cordless phones
Cruise control
Digital cameras
Disk drives
Electronic card readers
Electronic instruments
Factory control
Fax machines
Home security systems
Life-support systems
Medical testing systems

Microwave ovens
Modems
MPEG decoders
Network cards
Network switches
Pagers
Photocopiers
Point-of-sale systems
Portable video games
Printers
Satellite phones
Scanners
Speech recognizers
Stereo systems
Televisions
Temperature controllers
Theft tracking systems
VCR's, DVD players
Video game consoles
Washers and dryers

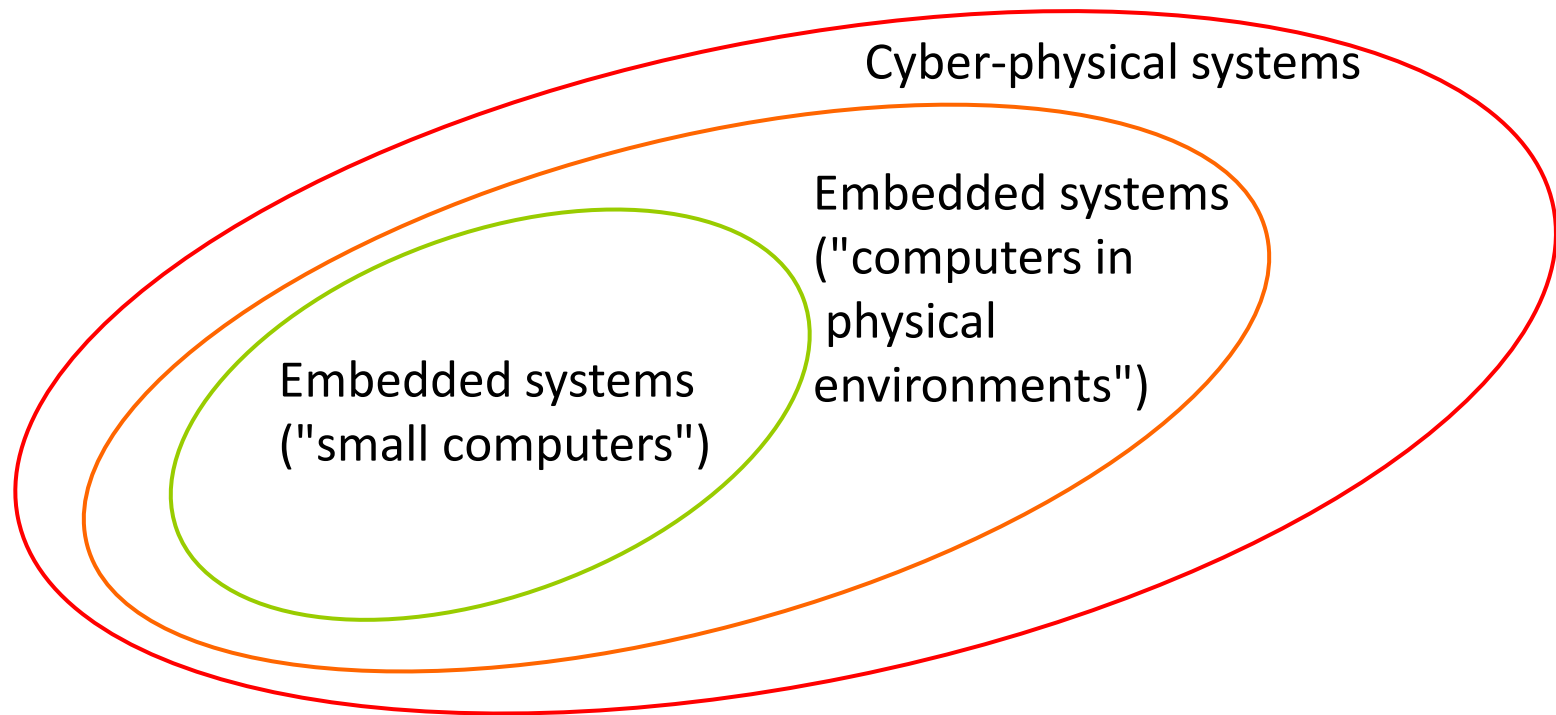


*What's your favourite
Embedded system??*

List in part obtained from: Vahid Givargis,
*Embedded Systems Design: A Unified
Hardware/Software Introduction*, 2000

Cyber-Physical Systems VS Embedded Systems

CPS = ES + physical environment



Deciding if it's an Embedded System

- The borders between *embedded and non-embedded computer system* are getting blurred...
 - Cellphones and PDAs ... embedded for not?
 - Set-top box for steaming internet TV?
- Traditional deciding factors are:
 - Task specific
 - Non-standard platform architecture
 - Limited resources

As more computing power moves into embedded systems, borders are getting more fuzzy.



To end off with...

Some class business

Prac Process & Assistance

Pracs done in a prac group of a two students.
Alternatively can be done individually.

Use Amathuba Discussions to save everyone time in answering questions and rebroadcasting answers. This way you could search the sites to get answers to problems much quicker instead of composing a new query.

NB: Pracs 1 and 2 are linked to the GA Test; it is important all students engage these pracs before writing the test as the GA Test questions probe understanding of techniques and practiced used in these pracs.

Prac Sessions this term ...?

Wednesday: Red Lab 16h00 to 18h00

Fridays: Red Lab 14h00 to 18h00

Tutors will be available in laboratory venues to help complete practicals*. Completion of practicals must be demonstrated to tutors on campus during these sessions. An announcement will be made to indicate when a new practical is released.

**More correctly to give you assistance and guidelines, not to finish pracs for you!*

Work packages: Tutorials and Practicals

Prac#	Covers
1	Timer Interrupts, Pushbuttons, LEDs, HAL (getting started / recapping STM development)
2	Pushbutton Interrupts, DAC, Filtering (Set up DAC, write function to generate signal)
3	ADC, PWM, SPI (SPI interfacing tasks)
4	Assembly Intro (Getting assembly module in program running)
5 *	Assembly Advanced (Assembly / C Integration and Interrupts)

* We aim to have this in, but if prac time and slots runs may be removed

Getting Ahead in EEE3096S



What **not** to do



- **Don't do Plagiarism, crime or other offenses**
 - The University takes plagiarism very seriously
- **Don't Let other students do your work**
 - Hand in only your own work for individual assignments
- **Don't do other students' work**
 - Group work must be divided up fairly
 - Keep track of each member's contribution

What you **can** do



- **Assist / suggest advice to other students**
 - Give procedural assistance in the labs
 - Refer other students to useful information and references
- **Good referencing practice**
 - Where you use somebody else's work, reference it properly. e.g. use IEEE referencing or Harvard referencing method

Working on Pracs

- **Think**

- Develop a plan of action, before you start hacking!

- **Try**

- Before saying that you don't know, try it!

- **Don't fry**

- Don't be careless and fry the equipment (or yourself!)



No frying or braaing in the lab!



Try to read Section 1 of Course Notes

and ...

get together a prac team

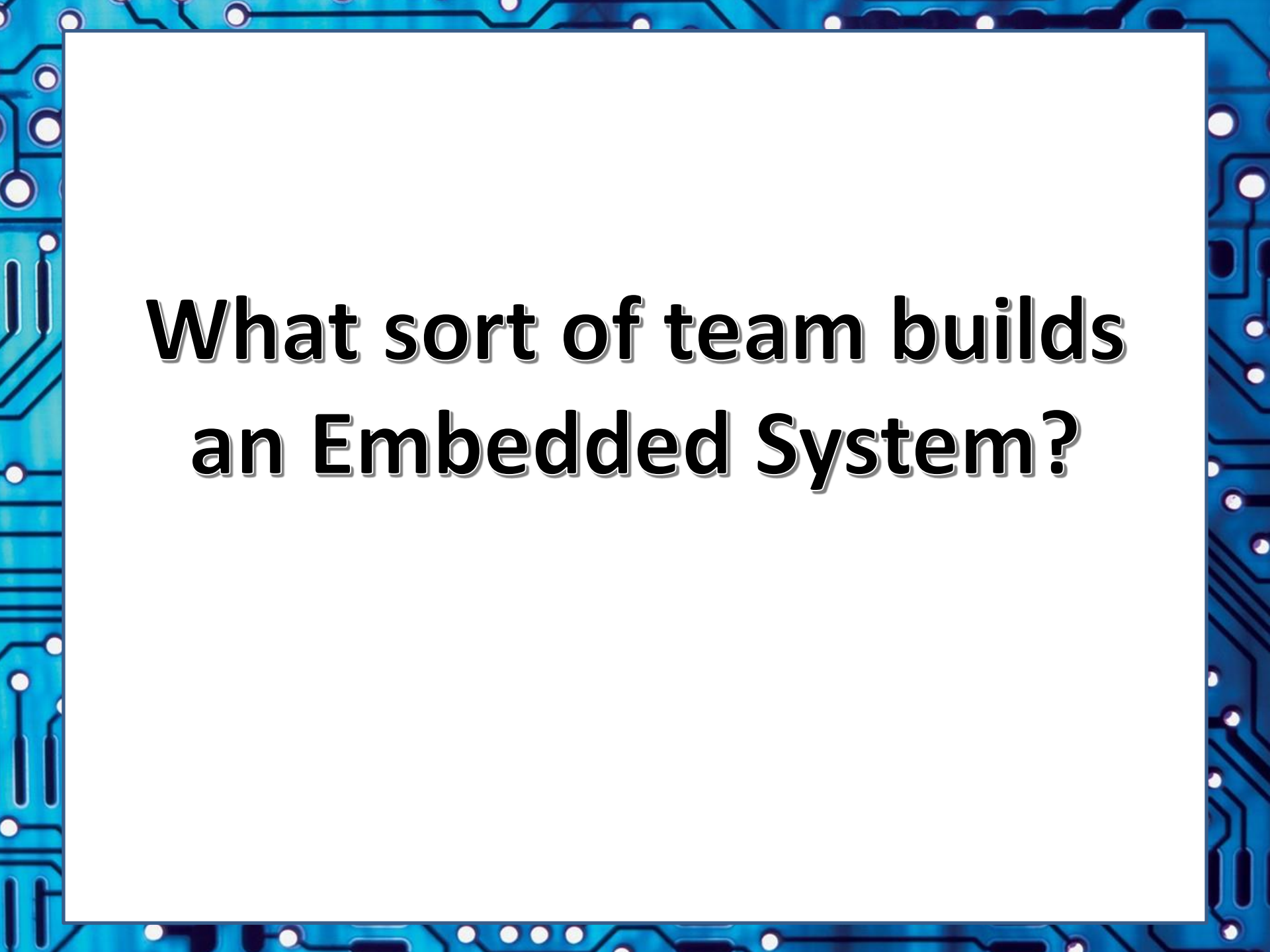
find your STM board

start on prac1

There is no prescribed textbook for this course, see last slide for recommended supplementary textbook.

Some additional suggested things to know (optional)

(ES professional should know this)

A decorative border with a blue and black circuit board pattern surrounds the central white text area.

**What sort of team builds
an Embedded System?**

Embedded System Team

- We'll go into more detail of this and process models later in the course... But here is a brief response to this question.
- Guiding point is: **“There's no a secret formula for a successful team”**
(Google did an [extensive study](#) on this)
- There are essentially **3 types of design teams**:
 - Centralized
Teams work as an agency where many projects may be going on and managers manage the workload of teams. Almost always, collaboration happens predominantly in the design team and the developers are shielded from other business process.
 - Embedded
Are multidisciplinary teams or Engineering, Product, and Design (EPD) teams. Cross-functionality teams consisting of about 7 members from different areas working for the same purpose in agile methodologies.
 - “Elevator team” or “Minimax team”
 - Can consider as an “on-demand team”. Designers temporary work into cross-functionality teams focused on a project and when the project completes, they return to the centralized team. [‘Minimax’ is similar, but referring to a minimum of the maximally skilled people for the task concerned in the hope it will be finished as quickly as possible.]

Additional reading (partly based on): <https://uxdesign.cc/3-design-team-structures-centralized-embedded-and-the-elevator-team-f2970008ed41>

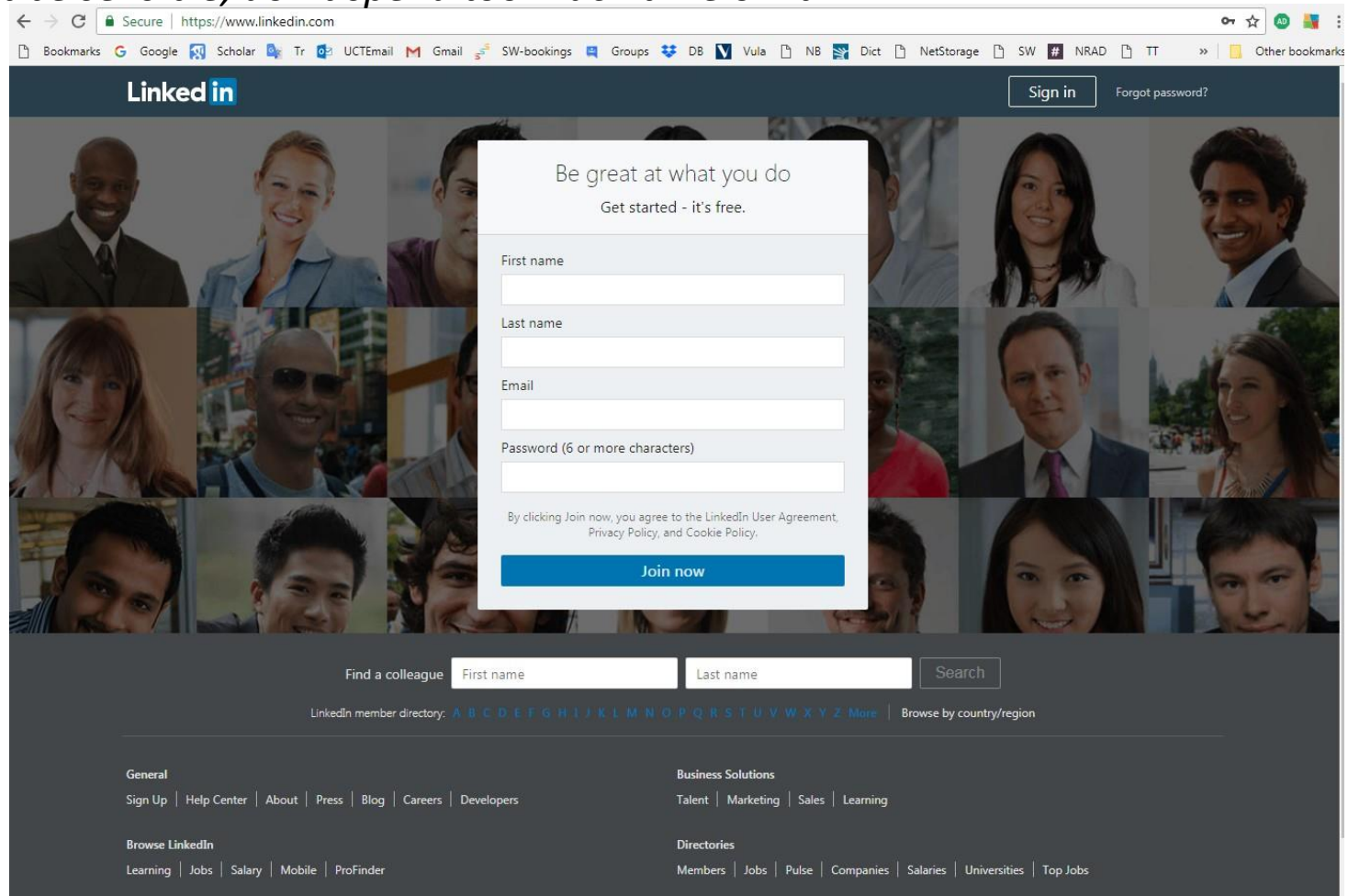
Embedded System Team

- Significant take-home point regarding ES teams
 - These often comprise a diversity of members from computer/mechatronic engineering and other disciplines such as 'domain experts'.
 - Embedded Systems engineers typically need an understanding of the context or 'plant' (as Americans tend to say) where the product will be used.
 - The 'plant' being, for example, the factory, work area or other physical context the product is to be used (this is usually a built environment as apposed to out in nature; but of course embedded systems are nowadays not limited to used in some sort of built environment).
 - A 'Domain Expert' refers to an expert that has a good understanding (or is a specialist) in the context or plant that the embedded system needs to work in and/or the type of operations that it needs to deliver. The domain experts are usually professionals from e.g. physics, healthcare, chemistry, etc. and advice on the needed formula, techniques etc that the embedded engineer needs to implement.

Recommended

(optional, for those who haven't already done this)

Consider building your own LinkedIn profile! or create one if you don't have.
But be sensible, don't spend too much time on it.



The screenshot shows the LinkedIn homepage with a central sign-up form. The form is titled "Be great at what you do" and "Get started - it's free." It contains four input fields: "First name", "Last name", "Email", and "Password (6 or more characters)". Below the fields is a blue "Join now" button. A disclaimer states: "By clicking Join now, you agree to the LinkedIn User Agreement, Privacy Policy, and Cookie Policy." The background features a grid of diverse professional headshots. At the top, there's a navigation bar with the LinkedIn logo, "Sign in", and "Forgot password?". Below the sign-up form is a search bar labeled "Find a colleague" with "First name" and "Last name" input fields and a "Search" button. At the bottom, there's a footer with links for "General" (Sign Up, Help Center, About, Press, Blog, Careers, Developers), "Business Solutions" (Talent, Marketing, Sales, Learning), "Browse LinkedIn" (Learning, Jobs, Salary, Mobile, ProFinder), and "Directories" (Members, Jobs, Pulse, Companies, Salaries, Universities, Top Jobs).

Secure | <https://www.linkedin.com>

Bookmarks | Google | Scholar | Tr | UCTEmail | Gmail | SW-bookings | Groups | DB | Vula | NB | Dict | NetStorage | SW | NRAD | TT | Other bookmarks

LinkedIn Sign in Forgot password?

Be great at what you do
Get started - it's free.

First name

Last name

Email

Password (6 or more characters)

By clicking Join now, you agree to the LinkedIn User Agreement, Privacy Policy, and Cookie Policy.

Join now

Find a colleague First name Last name Search

LinkedIn member directory: [A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#) [More](#) | [Browse by country/region](#)

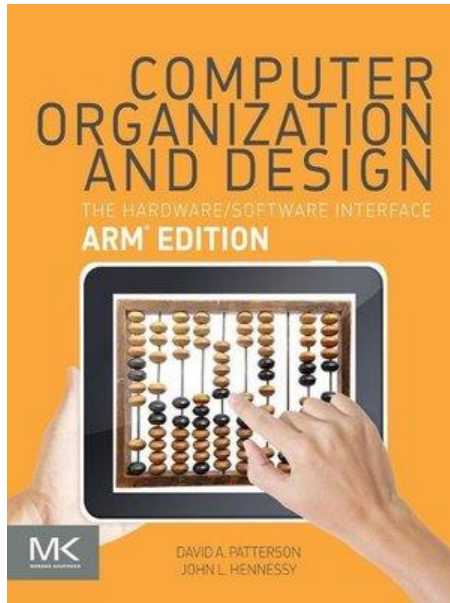
General
[Sign Up](#) | [Help Center](#) | [About](#) | [Press](#) | [Blog](#) | [Careers](#) | [Developers](#)

Business Solutions
[Talent](#) | [Marketing](#) | [Sales](#) | [Learning](#)

Browse LinkedIn
[Learning](#) | [Jobs](#) | [Salary](#) | [Mobile](#) | [ProFinder](#)

Directories
[Members](#) | [Jobs](#) | [Pulse](#) | [Companies](#) | [Salaries](#) | [Universities](#) | [Top Jobs](#)

Suggested Supplementary Textbook



Why this choice? It's probably obvious...

1. Because Prof. David Patterson (Uni. Of California, Berkeley) is a world renown expert in computer architecture and in CPS and Embedded Systems. Prof. John Hennessy (Stanford Uni.) is likewise highly distinguished particularly in his expertise and contributions to ARM and RISC processors.
2. It's chosen with the transferability of your (the student's) skills in mind... this book was chosen as it is designed around being globally relevant to this subject and well as relevant to both working in industry or pursuing further postgraduate study.

Recommended Textbook (not prescribed)

"Computer Organization and Design ARM Edition: The Hardware Software Interface" (The Morgan Kaufmann Series in Computer Architecture and Design) 1st Edition. Authors: David A. Patterson and John L. Hennessy.

<https://www.amazon.com/Computer-Organization-Design-ARM-Architecture/dp/0128017333>